Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 4$ in converting between fractions and decimals, evaluate $f(5)$.

Final Answer: 24. Substitute $x = 5$: $f(5) = 4(5) + 4 = 24$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 5$ in recurring decimals to fractions, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 5(2) + 5 = 15$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for recurring decimals to fractions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 6$ in multiplying with decimals, evaluate $f(2)$.

Final Answer: 18. Substitute $x = 2$: $f(2) = 6(2) + 6 = 18$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for multiplying with decimals consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 7$ in dividing with decimals, evaluate $f(2)$.

Final Answer: 11. Substitute $x = 2$: $f(2) = 2(2) + 7 = 11$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for dividing with decimals consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Coordinate Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Show that the points $A(1, 3)$, $B(4, 7)$ and $C(7, 3)$ form an isosceles triangle.

$AB = \sqrt{9 + 16} = 5$. $BC = \sqrt{9 + 16} = 5$. $AC = \sqrt{36 + 0} = 6$. Since $AB = BC$, triangle is isosceles.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Two similar cylinders have radii 4 cm and 6 cm. If the volume of the smaller is 160π cm³, find the volume of the larger.

$k = 6/4 = 1.5$. Volume ratio $= 1.5^3 = 3.375$. Volume $= 160\pi \times 3.375 = 540\pi$ cm³.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

If $f(x) = 4x - 3$ and $g(x) = x^2 + 1$, find (a) $gf(2)$ and (b) an expression for $g^{-1}(x)$ for $x \geq 1$.

(a) Inner function first: $f(2) = 4(2) - 3 = 5$. Then $g(5) = 5^2 + 1 = 26$. (b) Inverse of g: Let $y = x^2 + 1 \Rightarrow x^2 = y - 1 \Rightarrow x = \sqrt{y - 1}$. Thus $g^{-1}(x) = \sqrt{x - 1}$ for $x \geq 1$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Using the same bag (5 red, 3 blue), find the probability of drawing exactly one red ball.

$P(\text{RB}) + P(\text{BR}) = \frac{5}{8} \cdot \frac{3}{7} + \frac{3}{8} \cdot \frac{5}{7} = \frac{15}{56} + \frac{15}{56} = \frac{30}{56} = \frac{15}{28}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Functions

[Mensuration · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Number

[Number · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Sets & Venn Diagrams

[Sets & Venn Diagrams · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Algebra & Graphs

[Probability & Statistics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Mensuration

[Transformations & Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Geometry

[Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 9$ in decimals and place value, evaluate $f(2)$.

a. 13 ✓

b. 15

c. 4

d. 17

Substitute $x = 2$: $f(2) = 2(2) + 9 = 13$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 3$ in adding and subtracting decimals, evaluate $f(2)$.

a. 9 ✓

b. 12

c. 6

d. 13

Substitute $x = 2$: $f(2) = 3(2) + 3 = 9$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for decimals and place value consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for adding and subtracting decimals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the equation of the line through $(4, 1)$ and $(5, -7)$.

a. $y = -8x + 31$

b. $y = -8x + 35$

c. $y = -8x + 33$ ✓

d. $y = -8x - 33$

$m = \frac{-7-1}{5-4} = -8$. $y - 1 = -8(x - 4) \Rightarrow y = -8x + 33$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the gradient of the line joining $(1, 0)$ and $(3, 2)$.

a. 2

b. **1** ✓

c. 0

d. -1

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2-0}{3-1} = 1$

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A line has gradient 3. What is the gradient of a line perpendicular to it?

a. $2/3$

b. $-1/3$ ✓

c. $-13/3$

d. 3

Perpendicular gradients multiply to -1: $3 \times m_2 = -1 \Rightarrow m_2 = -1/3$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the distance between $(0, 0)$ and $(6, 8)$.

a. 3

b. **10** ✓

c. 6

d. 13

$d = \sqrt{(6-0)^2 + (8-0)^2} = \sqrt{100} = 10$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A line has gradient -5. What is the gradient of a line parallel to it?

a. -1 **b.** -5 ✓

c. -4 **d.** $1/5$

Parallel lines have equal gradients: -5.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the equation of the line with gradient 3 passing through $(-2, -3)$.

a. $y = 3x + 3$ ✓ **b.** $y = 3x + 5$

c. $y = 3x - 3$ **d.** $y = 3x + 1$

$y + 3 = 3(x + 2) \Rightarrow y = 3x + 3$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Area of trapezium with parallel sides 5 and 9, height 4:

a. **28** ✓ **b.** 56

c. 36 **d.** 14

Applying Geometry principles, $A = \frac{1}{2}(5 + 9)(4) = 28$ Therefore, the verified correct option is 28.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In a triangle, two angles are 68° and 67°. Find the third angle.

a. **45**$^\circ$ ✓ **b.** 52°

c. 57° **d.** 56°

Angles sum to 180°: third angle $= 180 - 68 - 67 = 45^\circ$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In a circle, a tangent meets the radius at:

a. 30° **b.** 45°

c. 60° **d.** **90°** ✓

Tangent is perpendicular to radius at point of contact.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Angle in a semicircle is always:

a. 45°

b. **90°** ✓

c. 180°

d. 60°

Thales' theorem: angle in semicircle = 90°.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A sector has radius 6 and angle 60°. Arc length is:

a. 2π ✓

b. 4π

c. π

d. 6π

$L = \frac{60}{360} \times 2\pi(6) = 2\pi$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Points P, Q, R lie on a circle with centre O. Tangents from external point T touch the circle at P and Q. If angle $\angle PTQ = 46^\circ$, what is the size of angle $\angle PRQ$ on the major arc?

a. 67° ✓

b. 74°

c. 134°

d. 92°

In quadrilateral $OPTQ$, angles $\angle OPT = \angle OQT = 90^\circ$ (tangents perpendicular to radius). The opposite angles sum to 180°: $\angle POQ = 180^\circ - 46^\circ = 134^\circ$. The angle at the circumference $\angle PRQ$ is half the angle at the centre: $\angle PRQ = \frac{134^\circ}{2} = 67^\circ$. [M1 for finding angle POQ = 134, M1 for dividing by 2, A1 for 67]

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Points A, B, and C lie on a circle with centre O. Angle $\angle AOB = 110^\circ$. What is the size of angle $\angle ACB$ on the major arc?

a. 55° ✓

b. 70°

c. 110°

d. 125°

By the circle theorem 'the angle at the centre is twice the angle at the circumference subtended by the same arc': $\angle ACB = \frac{1}{2}\angle AOB = \frac{110^\circ}{2} = 55^\circ$. [M1 for half angle at centre, A1 for 55]

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 2.01 Algebra & Graphs**

$ABCD$ is a cyclic quadrilateral. Angle $\angle DAB = 78^\circ$ and angle $\angle ABC = 104^\circ$. What is the size of angle $\angle BCD$?

a. 76°

b. 102° ✓

c. 104°

d. 78°

Opposite angles of a cyclic quadrilateral sum to 180°. Angle $\angle BCD$ is opposite to angle $\angle DAB$: $\angle BCD = 180^\circ - 78^\circ = 102^\circ$. [M1 for 180 - 78, A1 for 102]

39 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 4.01 Mensuration**

Two tangents TP and TQ are drawn from an external point T to a circle with centre O. If angle PTQ = 46°, find angle POQ.

a. **134° ✓**

b. 46°

c. 144°

d. 90°

Tangents are perpendicular to radii ($\angle OPT = \angle OQT = 90^\circ$). In quadrilateral OPTQ, opposite angles sum to 180°: $\angle POQ = 180^\circ - 46^\circ = 134^\circ$. [M1 A1]

40 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 5.01 Geometry**

Two chords AB and CD of a circle intersect at an interior point P. If $AP = 4$ cm, $PB = 9$ cm, and $CP = 3$ cm, what is the length of PD?

a. 10 cm

b. **12 cm ✓**

c. 14 cm

d. 16 cm

By the Intersecting Chords Theorem: $AP \times PB = CP \times PD \Rightarrow 4 \times 9 = 3 \times PD \Rightarrow 36 = 3 \times PD \Rightarrow PD = 12$ cm. [M1 for AP*PB = CP*PD, A1 for 12]

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

A line TA is tangent to a circle at A. ABC is a triangle inscribed in the circle. If angle $\angle TAB = 52^\circ$, what is the size of angle $\angle ACB$?

a. 38° **b.** 52° ✓

c. 104° **d.** 128°

By the Alternate Segment Theorem, the angle between a tangent and a chord is equal to the angle subtended by the chord in the alternate segment: $\angle ACB = \angle TAB = 52^\circ$. [B1]

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Transformations & Vectors

Two tangents TP and TQ are drawn from an external point T to a circle with centre O. If angle PTQ = 50°, find angle TPQ.

a. **65°** ✓ **b.** 50°

c. 75° **d.** 55°

Tangent lengths from a common external point are equal ($TP = TQ$). Triangle TPQ is isosceles with vertex angle 50°: $\angle TPQ = \frac{180^\circ - 50^\circ}{2} = 65^\circ$. [M1 A1]

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Probability & Statistics

What is the size of each exterior angle of a regular octagon (8-sided polygon)?

a. 45° ✓ **b.** 60°

c. 135° **d.** 30°

The sum of the exterior angles of any convex polygon is 360°. For a regular octagon with 8 equal sides: $\text{Exterior angle} = \frac{360^\circ}{8} = 45^\circ$. [B1]

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Functions

A rhombus has diagonals of lengths 10 cm and 24 cm. What is the perimeter of the rhombus?

a. 52 cm ✓

b. 48 cm

c. 68 cm

d. 34 cm

Diagonals bisect at right angles: half-diagonals are 5 cm and 12 cm. Side length = $\sqrt{5^2 + 12^2} = 13$ cm. Perimeter = $4 \times 13 = 52$ cm. [M1 A1]

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number

What is the sum of the interior angles of a convex hexagon (6-sided polygon)?

a. 360°

b. 540°

c. 720° ✓

d. 900°

Formula for sum of interior angles: $(n - 2) \times 180^\circ$. For $n = 6$: $(6 - 2) \times 180^\circ = 4 \times 180^\circ = 720^\circ$. [M1 for formula, A1 for 720]

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